Question:

Select a suitable software engineering model that aligns with the needs mentioned and justify your choice by explaining how the characteristics of the selected model support the constraints and goals of the project. Identify the key project management and development roles required in this model and describe the responsibility each role must demonstrate to deliver a scalable, maintainable, and high-quality software solution.

The case studies are given below:

Case Study 1

A medical device manufacturer is developing software for a **Smart Intensive Care Unit (ICU) Patient Monitoring System**. The software continuously monitors patients' heart rate, blood pressure, oxygen saturation, respiratory rate, and other vital signs. If abnormal values are detected, the system immediately alerts medical staff. Since the software directly supports life-critical healthcare decisions, even a small defect may put patients' lives at risk. Government regulations require complete documentation, strict verification, and validation of every system requirement before deployment. Each development activity must have a corresponding testing activity to ensure complete traceability between requirements and test cases. Once development starts, requirements rarely change because they are defined according to medical standards and regulatory compliance.

Case Study 2

A software company has been hired to develop a **Comprehensive University Management System** for a large university. The university administration wants students and faculty members to begin using essential features as early as possible instead of waiting for the entire system to be completed. The project includes several modules such as student admission, course registration, examination management, library services, hostel management, online fee payment, and alumni management. The university agrees that the software can be delivered module by module over several months, allowing users to start using completed features while additional modules are developed. Feedback from users after each release will be incorporated into future increments, and each completed module must integrate smoothly with the existing system.

Case Study 3

A government transportation authority plans to develop a **National Driving License Management System** to replace its existing paper-based licensing process. The system will manage citizen registration, learner permits, driving tests, license issuance, renewal, and violation records. Since the system must comply with existing government regulations and legal procedures, all functional and security requirements have already been finalized after several months of consultation with legal experts and policymakers. Any changes after development begins require lengthy administrative approval and are therefore highly discouraged. The project involves multiple government departments, requiring extensive documentation and formal approval at every stage before proceeding to the next phase. The client prefers a structured development process with comprehensive planning, detailed documentation, and complete testing before deployment. The system will only be released after all modules have been fully developed and verified.

Case Study 4

A startup company plans to develop an **AI-Powered Interior Design Application** that allows homeowners to visualize furniture arrangements, wall colors, lighting styles, and room decorations before making purchasing decisions. Although the client has a general idea of the desired application, they are unsure about the user interface, navigation flow, recommendation features, and visualization options. Potential customers are also expected to provide feedback throughout development regarding usability and design preferences. The company wants developers to build an early working version of the system so users can evaluate the interface and suggest improvements before the final software is developed. Multiple rounds of user feedback and requirement refinement are expected throughout the project.

Case Study 5

A software development company has received a contract to build a Smart City Citizen Service Portal within four months. The portal will provide multiple independent services, including online utility bill payment, waste collection requests, parking management, public transport information, complaint submission, emergency service requests, and community event management. Each service functions as a separate module with clearly defined requirements that have already been finalized for the initial release. The client expects rapid development with frequent demonstrations and prototype reviews to ensure that each module meets user expectations. Several experienced development teams will work simultaneously on different modules using reusable software components and visual development tools. Once individual modules are completed, they will be integrated into a single unified portal before deployment.

Example How to Answer the Questions:

Q1. A software company has been contracted to develop a **Smart Library Management System** for a large university. The university wants to modernize its library services by introducing digital book catalogs, online book reservations, borrowing and returning management, fine calculation, e-book access, and library analytics. However, the university administration does not want to wait until the entire system is completed before using it. Instead, they prefer that the software be delivered module by module so librarians and students can begin using completed features immediately. For example, the book catalog and membership management modules should be available first, followed by borrowing services, fine management, e-book access, and reporting modules. After each release, librarians will provide feedback, and improvements will be incorporated into future versions. The university also expects every newly developed module to integrate smoothly with the existing system without disrupting previously released functionality.

Select a suitable software engineering model that aligns with the needs mentioned and justify your choice by explaining how the characteristics of the selected model support the constraints and goals of the project. Identify the key project management and development roles required in this model and describe the responsibility each role must demonstrate to deliver a scalable, maintainable, and high-quality software solution.

Answer:

Selected Software Engineering Model is **Incremental Process Model**.

Justification:

The **Incremental Process Model** is the most suitable choice because the university wants the software to be delivered **module by module** rather than waiting for the complete system to be finished. This model allows essential features such as the digital catalog and membership management to be released first so that librarians and students can begin using the system immediately. Additional features, including borrowing management, fine calculation, e-book access, and analytics, can be added in later increments.

The case study also mentions that **feedback will be collected after each release**, allowing developers to improve future increments based on user experience. Furthermore, each newly developed module must integrate with the existing system, which is a key characteristic of the Incremental Model. Therefore, this model supports early software delivery, gradual feature expansion, and continuous improvement.

Project Roles and Responsibilities:

Role	Responsibility
Project Manager	Plans and monitors each increment, allocates resources, and manages the project schedule.
System Analyst	Collects and prioritizes requirements for each software increment.
Software Architect	Designs the overall system architecture to ensure smooth integration of future modules.
Developers	Design, code, and implement each increment according to the planned requirements.
Test Engineer	Tests every increment and verifies compatibility with previously released modules.
Librarian/User Representative	Evaluates each release, provides feedback, and validates whether the software meets user needs.

Q2. A startup company is developing an **AI-Based Personal Fitness and Nutrition Assistant** for smartphones. The application will recommend personalized workout plans, meal suggestions, progress tracking, and motivational coaching based on user preferences and health goals. Although the client has a general vision for the application, they are uncertain about the user interface, dashboard layout, navigation, visualization of fitness progress, and interaction with the AI assistant. Since user experience is considered the most important aspect of the application, fitness trainers and potential users will frequently evaluate early versions of the software and suggest changes to improve usability. The development team expects multiple rounds of interface redesign and feature refinement before the final product is implemented.

Select a suitable software engineering model that aligns with the needs mentioned and justify your choice by explaining how the characteristics of the selected model support the constraints and goals of the project. Identify the key project management and development roles required in this model and describe the responsibility each role must demonstrate to deliver a user-friendly, maintainable, and high-quality software solution.

Answer:

Selected Software Engineering Model is **Prototype Model**.

Justification:

The **Prototype Model** is the most appropriate choice because the client's requirements are **not completely defined** at the beginning of the project. The case study clearly states that the client is uncertain about the application's user interface, dashboard design, navigation, and interaction with the AI assistant. These uncertainties make it difficult to develop the final system directly.

The Prototype Model allows developers to build an initial working version of the application that users, fitness trainers, and stakeholders can evaluate. Based on their feedback, the prototype can be refined through several iterations until the requirements become clear. Since the project expects **multiple rounds of user feedback and interface improvements**, the Prototype Model provides an effective approach for developing a user-friendly application that satisfies customer expectations.

Project Roles and Responsibilities

Role	Responsibility
Project Manager	Coordinates project activities, schedules prototype development, and manages communication with stakeholders.
System Analyst	Gathers initial requirements and records customer feedback after each prototype evaluation.
UI/UX Designer	Designs user interfaces and improves usability based on customer suggestions.
Prototype Developer	Develops and modifies the prototype according to user feedback.
Test Engineer	Evaluates prototype functionality and identifies usability or functional issues.
Customer/Fitness Trainer	Reviews each prototype, provides suggestions, and validates whether the system meets user expectations.